Gesesew Reta Habtie

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Employment

2025 –	**Head of Department** of Physics, Debre Berhan University, DB, Ethiopia.
2024 –	**Assi. Prof.** Department of Physics, Debre Berhan University, DB, Ethiopia.
2016 – 2019	**Lecturer.** Department of Physics, Debre Berhan University, DB, Ethiopia.
2012 – 2016	**Lecturer.** Department of Physics, Bulehora University, BH, Ethiopia.

Education

2025	**Ph.D. University of Calcutta, India.** in Astrophysics. Thesis title: *Study of Novae.*
2015	**M.Sc. University of Ghana, Ghana.** in Mathematical Science. Thesis title: *The Stationary State of Magnetic field Near the Sun*.
2011	**M.Sc. Dilla University, Ethiopia.** in Physics. Thesis title: *Radiation Therapy*.
2009	**B.Ed. Debre Berhan University, Ethiopia.** in Physics. Thesis title: *Attitude of Students Towards Physics*.

Skills

Proposals & Observations	I have successfully submitted and been allotted over 80 hours of observation time on the 2 m Himalayan Chandra Telescope (HCT). During this period, I observed various types of novae—including dwarf and symbiotic novae—using the **HFOSC** (Gr7&8; *BVRI* filters), **TIRSPEC** (*JHK* filters), and **HESP** (R = 30,000) instruments.
Data Reduction	IRAF, PyRAF, Python, Astropy.
Programing Languages	Python, Fortran, awk, and Shell scripting.
Modelling	Cloudy, Shape, fityk, and PyCloudy.
Operating System	Linux, Mac, and MS. Window.
Typesetting tools	Latex, MS Office (word & excel), html.
Languages	Amharic (**Native**), English (**Fluent**), and Ge'ez (**Intermediate**).

Talk & Poster Presentations

Jan. 2026	College Seminar, 2026, Debre Berhan University, Ethiopia (**Talk**).
Oct. 2025	11^{th} Meeting of **BRICS Astronomy Working Group (BAWG) 2025**, Brazil (**Poster**).
Dec. 2024	College Seminar, 2024, Debre Berhan University, Ethiopia (**Talk**).
June 2024	Symbiotic stars, weird novae, and related embarrassing binaries conference, Charles University, Prague (**Talk**).
Feb. 2024	SNBNCBS Bose fest, SNBNCBS, Kolkata, India (**Talk**).
Jan. 2023	42^{nd} annual meeting of the Astronomical Society of India, IISc, Bengaluru, India (**Talk**).
Mar. 2022	40^{th} annual meeting of the Astronomical Society of India, IIT Roorkee, Roorkee, India (**Poster**).

Workshops and Summer School attended

Apr. 2026	20^{th} General Assembly of ESSS, Addis Ababa, Ethiopia.
Oct. 2025	Virtual Observatory Workshop on the **BRICS Astronomy Working Group (BAWG) 2025**, Sao Paulo, Brazil.
Apr. 2025	20^{th} General Assembly of ESSS, Addis Ababa, Ethiopia.
Aug. 2024	32^{nd} IAU General Assembly, Cape Town, South Africa.
Jan. 2024	4^{th} Meeting on Star Formation Star Formation Studies in India, SNBNCBS, Kolkata, India.
Jan. 2023	UV-Optical-IR Astronomy in India: Prospects of the next decade, IISc, Bengaluru, India.
Mar. 2018	40^{th} International School for Young Astronomers, Cairo university, Egypt.
June 2015	Computing in Applied Mathematics (CAMS) workshop, AIMS Ghana, Ghana.

Awards

2025	Selected for the C2F Postdoctoral Fellowship (offer not taken).
2019	Selected for the TWAS-Bose Ph.D. Fellowship.
2017	Awarded second prize in NASA's CCMC Student Research Contest.
2014	Selected for the AIMS Ghana M.Sc. Fellowship.

Research Grant

2019	won 30,000 ETB to conduct a research on "*Investigation of Indoor Radon Concentration in the Dwelling areas of Debre Berhan Town, Ethiopia*".
2026	won 380,000 ETB to conduct a research on "*Documenting and Investigating Indigenous Knowledge of Calendars, Astronomy, and Prediction in North Shoa Zone, Ethiopia.*".

Teaching

2025 –	Space Science. Debre Berhan University, DB, Ethiopia.
2024 – 2025	General Physics. Debre Berhan University, DB, Ethiopia.
2016 – 2019	Introduction to Astronomy, Stellar Physics, Nuclear Physics, Electrodynamics, Mechanics and Heat, Electromagnetism, and Modern Physics. Debre Berhan University, DB, Ethiopia.
2012 – 2014	Mathematical Methods of Physics, Medical Physics, Radiaton Physics, Mechanics and Heat, Electromagnetism, and Modern Physics. Bule Hora University, BH, Ethiopia.

References

1. **Dr. Ramkrishna Das,** *SNBNCBS, JD Block, Sector III, Salt Lake, 700106, Kolkata, India.*
   Email: ramkrishna.das@bose.res.in
2. **Dr. Elias Aydi,** *Texas Tech University, Box 41051, Lubbock, TX, 79409-1051, USA.*
   Email: eaydi@ttu.edu
3. **Prof. N. M. Ashok,** *PRL, Navrangpura, Ahmedabad 380 009, Gujarat, India.*
   Email: ashoknagarhalli@gmail.com
4. **Dr. Tapas Baug,** *SNBNCBS, JD Block, Sector III, Salt Lake, 700106, Kolkata, India.*
   Email: tapasbaug@bose.res.in

Research Publications

Refereed Journal Articles

1. **G. R. Habtie**, “3d morpho-kinematic study of nova v1405 cas (2021),” 2026 (To be Submitted).
2. **G. R. Habtie**, R. Das, E. Aydi, and P. A. Dubovsky, “Optical spectroscopy and temporal evolution of the nova v1405 cas,” *arXiv e-prints*, arXiv:2607.19138, 2026. DOI: 10.48550/arXiv.2607.19138. arXiv: 2607.19138 [astro-ph.SR].
3. E. Aydi, J. D. Monnier, A. Mérand, *et al.*, “Multiple outflows and delayed ejections revealed by early imaging of novae,” *Nature Astronomy*, 2025. DOI: 10.1038/s41550-025-02725-1.
4. **G. R. Habtie** and R. Das, “Spectroscopic study of the quiescent stages in between the 2006 and 2021 outbursts of rs ophiuchi,” *Monthly Notices of the Royal Astronomical Society*, vol. 537, no. 2, pp. 2046–2060, 2025. DOI: 10.1093/mnras/staf119. arXiv: 2402.03234 [astro-ph.SR].
5. **G. R. Habtie**, R. Das, R. Pandey, N. M. Ashok, and P. A. Dubovsky, “Correction to: Study of the fastest classical nova, V1674 Her: photoionization and morpho-kinemetic model analysis,” *MNRAS*, vol. 529, no. 2, pp. 917–917, Apr. 2024. DOI: 10.1093/mnras/stae617.
6. **G. R. Habtie**, R. Das, R. Pandey, N. M. Ashok, and P. A. Dubovsky, “Study of the fastest classical nova, v1674 her: Photoionization and morpho-kinematic model analysis,” *Monthly Notices of the Royal Astronomical Society*, vol. 527, no. 1, pp. 1405–1423, 2024. DOI: 10.1093/mnras/stad3295. arXiv: 2310.16619 [astro-ph.SR].
7. R. Pandey, **G. R. Habtie**, R. Bandyopadhyay, R. Das, F. Teyssier, and J. Guarro Fló, “Study of 2021 outburst of the recurrent nova rs ophiuchi: Photoionization and morpho-kinematic modelling,” *Monthly Notices of the Royal Astronomical Society*, vol. 515, no. 3, pp. 4655–4668, 2022. DOI: 10.1093/mnras/stac2079.

Conference Proceedings

1. **G. R. Habtie**, R. Das, R. Pandey, N. M. Ashok, and P. A. Dubovsky, “Study of the fastest classical nova, v1674 her: Photoionization and morpho-kinematic model analysis,” in *42nd Meeting of the Astronomical Society of India (ASI)*, vol. 42, 2024, O51, O51.
2. **G. R. Habtie** and T. del Rio Gaztelurrutia, “The stationary state of magnetic field near the sun,” in *Solar Heliospheric and Interplanetary Environment (SHINE 2017)*, 2017, 170, p. 170.

Astronomer’s Telegrams / Non-Peer-Reviewed

1. **G. R. Habtie**, R. Das, and R. Pandey, *T crb on the verge of an outburst: $H\alpha$ profile evolution and accretion activity*, The Astronomer’s Telegram, No. 17041, 2025.
2. **G. R. Habtie** and R. Das, *He ii emission line became the most intense line in the optical spectrum of nova v1405 cas*, The Astronomer’s Telegram, No. 16496, 2024.
3. **G. R. Habtie** and R. Das, *Optical photometry and spectroscopy of nova sco 2024 (v1723 sco)*, The Astronomer’s Telegram, No. 16454, 2024.